AI Alignment and Governance as Normative Institutional Design

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Technical approaches to AI alignment and governance have increasingly drawn on traditions developed for problems in moral philosophy, political theory and legal systems—from constitutionalism and social choice, to deliberative democracy and virtue ethics. However, technical alignment methods draw on these traditions unevenly. Some traditions, such as constitutionalism, have inspired substantial bodies of technical work, while others remain largely unexplored. Moreover, alignment methods often carry over only a fraction of the concepts that make up the traditions they draw on. In order to provide a systematic understanding of this landscape, we construct a two-dimensional property space that places AI alignment and governance approaches into fifteen regimes, and maps each approach to the traditions in philosophy, political theory, and legal theory that it draws on. We use this property space to study *normative translation*—how methods carry over the concepts of the traditions they draw on—analyzing which concepts are preserved, partially preserved, or omitted, and what consequences these choices bring about. We highlight a recurring tendency of alignment methods to treat normative disagreement as a problem to be *resolved away*, often despite optimizing for incommensurable objectives and drawing from underlying traditions that argue against it. Finally, mapping the full landscape reveals underexplored regimes and omitted concepts. We argue that these gaps point toward a different goal for AI alignment: not invisibly collapsing disagreement, but building methods that can hold and contest it.

Keywords: AI Alignment, AI Governance, Moral Philosophy, Political Theory, Legal Theory, Value Pluralism, Institutional Design

AI Alignment & Governance

1. Introduction

What should an AI system do when the values it is expected to follow conflict? How should human preferences that differ across individuals determine its behavior? Should those preferences be aggregated into an objective, encoded as explicit principles, or resolved through deliberation? And when reasonable people continue to disagree, should an aligned system resolve it at all, or instead preserve and govern disagreement over time?

These questions are increasingly central to AI alignment, but they are not uniquely questions about AI. Subfields of philosophy have long studied how competing values and obligations should be weighed (Berlin, 1969; Ross, 1930); political theory has asked how institutions can govern populations that disagree about what is good or just in a legitimate and stable fashion (Habermas, 1996; Rawls, 1993); and legal jurisprudence has developed systems for specifying and contesting norms over time, as societies evolve (Dworkin, 1988; Hart, 1961). These traditions have been realized in concrete institutions: from codified rules in constitutions to democratic procedures aggregating competing preferences; resulting in different institutional arrangements for operating and governing under disagreement. In recent years, the fields of AI alignment and governance have begun to draw on these traditions. For example, Constitutional AI (Bai et al., 2022) encode normative principles in an explicit artifact written in natural language that humans can understand and AI systems are expected to follow (often called "a constitution", "model spec", or "code of conduct"); social-choice approaches to preference learning aggregate judgments according to individuals' preferences or incentives (Ge et al., 2024; Maura-Rivero et al., 2025); deliberative

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approaches derive principles through collective procedures (Huang et al., 2024); and pluralistic systems attempt to represent information and ultimately respond in line with heterogeneous and sometimes directly opposing values (Feng et al., 2024; Sorensen et al., 2025). And as a result of the success of these technical approaches, concepts originating in moral philosophy, jurisprudence, and political theory have become a regular part of the technical vocabulary of AI alignment and governance.

Yet, this transfer of conceptual methods is often selective. For example, Constitutional AI (Bai et al., 2022) draws on constitutionalism but carries over only some of its concepts, as we illustrate in Section 4. Such simplifications may be products of choices made due to engineering or infrastructural constraints, but often they represent design choices, and are ones that are crucial to the resulting system. Our broader concern with such transfer is that the **normative behavior of a system can change when a concept is separated from the institutional context in which it performed its function**. We call the process by which a method carries over the concepts of a tradition *normative translation*, and in this paper, ask what consequences follow when concepts are only partially preserved in translation.

In order to study this, we systematically map AI alignment and governance approaches to the traditions in political theory, legal theory, and philosophy that they draw on, by asking which regime each approach falls in. Following Collier et al. (2012)'s typology construction, we organize approaches according to two recurring dimensions: where normative authority resides and what happens when norms conflict. Mapping existing approaches into this space reveals an uneven landscape: while alignment systems increasingly incorporate heterogeneous sources of normative input, comparatively few preserve disagreement as an enduring feature of the resulting system. This allows us to understand not only which normative concepts have influenced technical alignment systems, but also to understand both existing translations of traditions and regions of the design space that remain technically underexplored.

Our contribution here is threefold. First, we survey AI alignment and governance approaches that draw on normative traditions. We provide a searchable encyclopedia of these traditions, their concepts, and the approaches that draw on them to aid researchers in work involving these concepts. Second, we construct a property space that organizes these approaches into regimes along two dimensions: the locus of normative authority and the disposition of normative conflict. This, in turn, allows us to discuss the *coverage* of the space. Third, we analyze how existing alignment methods translate the concepts of the traditions they draw on, distinguishing shortfalls against concepts a method claims from omissions of concepts it never set out to carry over, and the potential consequences of each. Our goal is not necessarily to critique AI systems for not reproducing a tradition identically, but rather, to enforce that drawing on a tradition should make its institutional assumptions explicit. By understanding the space of concepts holistically, and identifying the gaps in current alignment methods, we hope to identify both the strengths and limitations of existing approaches, as well as provide a systematic basis for exploring alternatives for future systems.

2. Related Work

Normative theory in AI alignment. A growing body of work draws explicitly on normative concepts from philosophy, political theory, and jurisprudence. Examples include pluralistic alignment (Sorensen et al., 2024), social choice for aggregating heterogeneous human preferences (**?**), deliberative democracy for collective alignment (Huang et al., 2024; Ovadya et al., 2024), constitutionalism for principle-based alignment (Bai et al., 2022), and jurisprudential concepts such as precedent and legal interpretation for case-based alignment and specification design (Caputo, 2024; Kolt et al., 2026). These works establish individual connections between normative traditions and technical mechanisms; we provide a broader mapping of these connections in Appendix A.

Typologies and property spaces in political and social sciences. Our approach builds on a longstanding tradition of typological analysis in political science and the social sciences, where multidimensional property spaces are used to form concepts, distinguish theoretically meaningful types, and expose combinations not captured by existing classifications (Barton, 1955; Collier et al., 2012; Elman, 2005; George and Bennett, 2005). This tradition decomposes variation into underlying dimensions whose cross product constitutes a property space. Importantly, cells in this property space, which are theoretically possible but unoccupied, can themselves be analytically informative.

Existing surveys of AI alignment. Recent surveys of AI alignment are primarily focused on individual subcategories of technical approaches (e.g., value pluralism) and accordingly organize systems according to technical mechanisms. For example, Sorensen et al. (2024) distinguish Overton, steerable, and distributional pluralism, while other frameworks distinguish approaches according to how preferences or perspectives are represented and aggregated. We instead construct a *property space* (Collier et al., 2012) from recurring dimensions of normative institutions, described next.

3. A Normative Design Space for AI Alignment and Governance

In order to study how traditions are drawn on across fields, we create a two-dimensional property space defined by two dimensions: the locus of normative authority and the disposition of normative conflict. We derive these dimensions through a systematic mapping of AI alignment and governance literature followed by conceptual substruction (Barton, 1955): "the procedure of finding, for a given system of types, the attribute space in which it belongs and the reduction which has been implicitly used" (Lazarsfeld, 1937).[1] Overlapping surface categories are, therefore, decomposed into the underlying attributes that distinguish them.

Unlike conventional typologies that discard unrealized combinations, we retain empty cells because absence from the technical literature is itself analytically informative for identifying underexplored architectures. Our approach also differs from existing taxonomies of AI alignment, which group approaches according to their surface-level technical mechanisms within tradition-derived categories, in that it allows methods from different traditions to be compared.

The construction is executed as follows. Per Petersen et al. (2008), two coders independently extract normative concepts from the sampled literature, reconcile near-equivalent categories and extract keywords that represent the respective concept. Keywords are unified across studies and the resulting clusters form categories of the scheme. Full collection and inclusion criteria, as well as statistics, are provided in Appendix B.

This yields two dimensions: the **locus of normative authority**, which indicates where the norms that bind a decision reside prior to the moment of decision (not who holds authority over them), and the **disposition of normative conflict**, which indicates what happens when those norms conflict—whether the conflict is resolved, kept open, or routed around. Their cross product gives a 5 x 3 property space, whose fifteen cells we call *regimes*. We define the terms used throughout the paper in Appendix C.

3.1. Locus of Normative Authority

The *locus* of a norm is where it resides before the decision it binds is made. For the locus of normative authority, we find the following categories:

1. **Codified.** Norms are stated as explicit rules in a written specification that is set before any particular case arises (Hart, 1961).
2. **Case-accumulated.** Norms reside in a growing body of past decided cases. No case is decided by a rule stated in advance, but by what accumulated decisions have held (Jon-

[1] We note a difference from the null resolution procedure of prior work (Lazarsfeld, 1937)—whereas that procedure removes any null cells, we retain them, since we are interested in gaps in the literature.

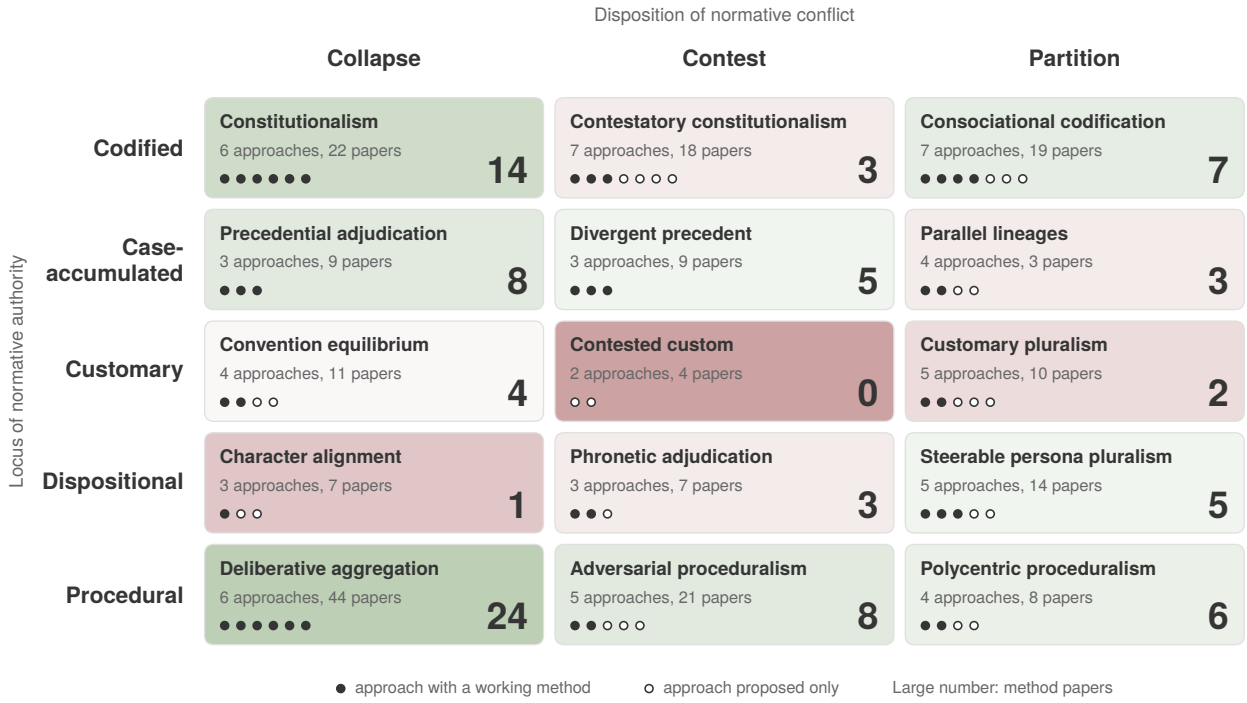


Figure 1 | **The Normative Design Space for AI Alignment and Governance.** The two axes, *locus of normative authority* (rows) and *disposition of normative conflict* (columns), yield fifteen *regimes* (cells). Each cell contains the respective regime's name, and counts of approaches and papers (within AI literature) that fall under it. Each dot below these statistics corresponds to one approach—filled where at least one working method exists, empty where the approach has only been proposed. The large number in the bottom right corner of each cell counts method papers, which is a subset of papers that explicitly specify a technical mechanism. The background color corresponds to the log of method paper count.

sen and Toulmin, 1988; Levi, 2013).

3. **Customary.** Norms reside in regularities of practice. Unlike codified or case-accumulated norms, they are never authoritatively stated; they are learned through participation and enforced by expectation rather than by rule or ruling (Burke, 1986; Lewis, 2008).
4. **Dispositional.** Norms reside in the system's character rather than in any external artifact. There is no rule to consult and nothing to look up, only evidence of character traits of the system itself (Aristotle).
5. **Procedural.** Norms do not persist between decisions. What should be done is whatever a specified procedure (e.g., a vote, a deliberation, a debate) yields at the moment of decision (Habermas, 1996).

3.2. Disposition of Normative Conflict

A *normative conflict* arises when two or more norms that apply to a case recommend incompatible actions. For the disposition of normative conflict, we find the following categories[2]:

1. **Collapse.** The conflict is settled by a ranking (e.g., a scalar objective, a weighted sum, or a priority order among principles) and the losing norms have no further force (i.e., nothing lets them reopen the decision or bear on later ones).[3]

[2]We considered open, axial and selective coding (?), as well as the first-order, second-order and aggregate-dimension ladder (Gioia et al., 2013), but note that the keyword-based approach is more native to the computer science literature it addresses. Alternatives would likely yield a comparable outcome in terms of the intermediate categories.

[3]Practically, most existing systems that fit this category

2. **Contest.** The conflict is preserved and the losing norms retain force. This means that the output either holds the competing norms open for a later decision (e.g., by setting out the positions, not choosing among them, and deferring to the user) or presents a standing channel (e.g., a mechanism for appeal or amendment), based on which the case itself can be reopened or have an impact on later ones (Mouffe, 2000; Pettit, 1997).
3. **Partition.** The conflict is avoided. Several normative orders are kept intact, each authoritative within its own domain, and a rule sorts cases among them by features of the case (e.g., who is asking, where, or in which domain) (Lijphart, 1977; Savigny, 1867; **?**).[4]

The categories are separated by two questions. First, is the conflict sorted to one of several normative orders by features of the case, without ranking the orders? If so, the disposition is partition. Otherwise, once the system acts, can the losing norm still change any one of the following: a decision the output leaves open, the decision itself on appeal, or later decisions of the system? If so, the disposition is contest; if not, collapse. We code dispositions at the level of the whole method, across both training and inference, since force is often removed in inference. For instance, a reward model utilizes minority judgments at training time but absorbs them into a single score at inference time.

Disposition and incommensurability. The three disposition categories differ in what they presuppose about the ability to rank conflicting norms against one another. Two values are *incommensurable* when no common measure ranks their realizations; neither option is better, yet they are not equally good, as a small improvement to one would not necessarily make it better than the other (**?**; cf. **?**, who reads the same pattern as parity rather than incomparability; see **?** for a more thorough overview).[5] Collapse presupposes a complete ordering of the options—whether achieved through a weighted sum of component values (see **?** for a critique) or a strict priority among principles—under which the rejected options are plainly worse. Where values are incommensurable, no such ordering exists. And so, if a decision is still made, it leaves a remainder (**??**), a loss that the chosen option does not cancel and that collapse erases. Contest and partition draw on two prominent responses to such value conflict in the pluralist tradition. Contest treats this conflict as a permanent condition to be governed rather than settled once and for all (Berlin, 1969; Mouffe, 2000; **?**). Partition rejects the comparison altogether, applying different decisions in different spheres, governed by their own norms (**??**).

Figure 2 traces how the method literature in each row and column of the space has accumulated over time. Figure 3 breaks each regime's literature down by the kind of contribution it makes.

[**roma:** let's add this in a limitations section] General morphological analysis (Ritchey, 2022) would be inappropriate here, given that it constructs its multi-dimensional space while inspecting unrealized combinations, which requires preemptive stipulation. Since we are working with an existing structure and literature landscape, this would introduce unnecessary assumptions and likely lead to a less grounded result.

4. Case Studies of Normative Translation in Alignment

To understand the translation of normative concepts into technical systems we use three representative alignment methods as case studies. These are chosen by virtue of their wide adop-

resolve the conflicting norms into a single output, without trace or acknowledgment of the disagreement that produced it. A system that acknowledges them can still collapse if the acknowledgment is not paired with potential further force ascribed to the losing norm.

[4]Partition does not eliminate conflict but relocates it to cases that fall between orders, where the sorting rule must decide which order governs, a problem known in law as choice of law (Savigny, 1867) (cf. de Sousa Santos, 2002 on interlegality).

[5]We use *incommensurable* for abstract values (e.g., helpfulness and harmlessness) and *incomparable* for the options that bear them (e.g., two candidate responses). Usage in the literature varies (Chang, 1997; **?**). Incommensurability is one ground for resisting collapse; reasonable disagreement among persons, which applies to both commensurable and incommensurable values, is another (Rawls, 1993).

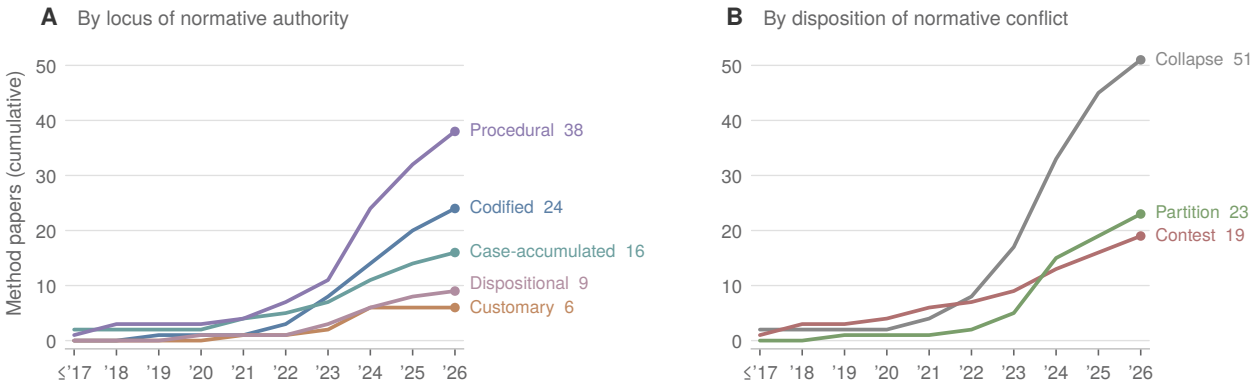


Figure 2 | **Growth of the Method Literature.** Cumulative number of method papers (in AI alignment and governance literature) by year of first publication, grouped (A) by locus of normative authority and (B) by disposition of normative conflict. Work published before 2018 is pooled into the first year.

tion as alignment techniques. For each approach, we examine (a) it's normative regime within our taxonomy, (b) the traditions it draws on, and the concepts that make up these traditions, and (c) whether the approach preserves, partially preserves, or omits concepts and (d) if consequences of these choices occur for the resulting normative system. We ensure that we distinguish connections claimed by the work itself, from those identified through our analysis.

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4.1. Constitutional AI and Constitutionalism

Constitutional AI (Bai et al., 2022) and adjacent methods that stem from it draw on the normative concept of **constitutionalism**. An explicit set of principles that a model should follow are codified in a unified artifact—often called the constitution or model spec, and used to guide or evaluate model behavior. This method instantiates **codified authority** with **collapsing conflict**, in that the normative specifications are codified before training as an authoritative source of guidance, that is then used to influence models' actions. In legal and political theory however, constitutionalism involves the codification of rules, tightly coupled with frameworks to interpret, apply, contest and revise these rules (Dworkin, 1988; Hart, 1961). The AI alignment version therefore preserves the first part of explicit codification, but only partially preserves interpretation of the rules: models apply the codified rules in different contexts without clear interpretation guidelines. Moreover, revising these rules do not exist as institutional frameworks: developers revise specifications before new model releases, but not in a procedural fashion, with mechanisms for precedent, stakeholder or public appeal, as well as oversight from external authorities. Lastly, frameworks for contesting these rules do not exist at all. The consequence of this is that the resulting normative structure does not equal constitutional governance as defined by the term. Merely employing a fixed specification largely separates the normative text from this surrounding institutional apparatus. This translation does preserve the constitution as a source of principles without preserving the mechanisms through which constitutional authority should be exercised, challenged or amended.

We are not suggesting that this translation is necessarily defective. Explicit specification of norms in constitutions makes normative guidance understandable by humans[6], and usable within existing training pipelines. But it also changes

[6]The term constitution also makes the concept more familiar to laypeople. This term also comes with an arguably favorable connotation. While this is positive for engaging laypeople in the process of AI governance, it can also establish a false sense that AI constitutionalism affords similar transparency and mechanisms of political constitutions, which is misleading.[**roma:** i think we should put this footnote in text! it seems important enough that it's not footnote material :)]

how normative authority is structured. Authority that is distributed across text, interpretation, precedent, and institutional roles becomes concentrated in the specification itself and in the process that authors it[7].

4.2. Preference-based Alignment and Utilitarianism

Preference-based alignment uses human judgements to train reward models that influence the behavior of AI systems by steering outputs towards approximated aggregate preferences (Christiano et al., 2017; Ouyang et al., 2022). Most widely used approaches instantiate **procedural authority** with **collapsing conflict**: judgements from populations of individuals are aggregated during training time, often masking individual or minority preferences. Conceptually, these ideas have roots in moral philosophy relating to utilitarianism, welfare and social good Bentham (1994); Mill (2011), although most technical preference optimization frameworks do not claim to implement these traditions. The technical translation therefore preserves individual judgements at input time: comparative judgments from multiple humans are preserved in the data,[8] but they ultimately contribute to a common reward function, rank ordering, or model policy—thus only representing the aggregate population vote. These methods therefore make a design choice to represent plurality upstream (data collection and training) but then collapse it downstream at inference time. This therefore omits procedures to determine how preferences should be weighted or how affected parties could challenge the objectives learned. It also does not preserve utilitarian concepts of maximizing welfare.

A clear consequence of this translation is that allowing for plurality of judgements at the source might not result in plurality in the resulting system. The design choices made to allow preference optimization models to work well at a large scale result in the resulting policy resolving diverse preferences into aggregates. This aggregation, while important for computational purposes, is a consequential normative choice about representing, preserving and weighting preferences of a population.

4.3. Collective Deliberation and Deliberative Democracy

Deliberative alignment systems explicitly incorporate people's input as a form of democratic participant to construct model specifications or influence alignment training Huang et al. (2024). Such approaches instantiate **procedural authority** with either **contest** or **collapsing** conflict—although multiple viewpoints are taken into account at different parts of the process, the resulting system often consolidates them in the final stages. These methods stem from theory in deliberative democracy that calls for the opportunity of individuals to participate in decision making, reasoning and challenging of collective goals (Habermas, 1996). Technical implementations of this do preserve collective judgements and deliberative traces, with some frameworks enforcing participants to interact and debate their viewpoints. However, institutional frameworks to guarantee representative participation, as well as continued processes to allow for subsequence contesting, or deliberating at multiple stages do not exist.

As originally intended by deliberative democracy frameworks, this translation does not allow for continual authority of collective judgement to exist. Alignment systems that follow this therefore take in collective public perspectives in one phase of the pipeline, but then consolidate these perspectives to operate under a unified policy in the resulting stages. Deliberative alignment systems can therefore result in architectures that collapse multiple viewpoints rather than preserving contestation. Thus, labeling a system "deliberative" is insufficient to determine its regime: the same procedure can instantiate different dispositions of normative conflict. This illustrates why

[7]The process of authoring constitutions is currently left largely up to the discretion of its creators, who may but are not required to consult other stakeholders. To be the best of our knowledge, there is currently no major model employing AI constitutionalism for its alignment that would enable contesting or directly amending the constitution for individuals outside of the authoring organization.

[8]Concretely, the data are relative judgments, collected by annotators who are shown two or more model responses and asked which they prefer or to rank them (Christiano et al., 2017; Ouyang et al., 2022).

the two dimensions of authority and conflict must be analyzed separately. Collective participation of individuals does not, in of itself, determine how disagreement is governed. Normative theory makes these combinations visible even where current technical implementations concentrate on only a subset of them.

4.4. The Consequence of Translation Gaps

[maty: todo - connections to rob's memo]

Across these examples, we see that normative translation tends to preserve the concept of a tradition that can most readily be represented as a technical mechanism: constitutions become specifications, heterogeneous preferences become aggregation problems, and deliberation becomes a procedure for producing judgments. The surrounding concepts through which institutions manage interpretation, disagreement, revision, and competing authority are less consistently preserved. However, as we have seen, this simplification of the concept into only the technical mechanism without the surrounding institutional structure shapes the normative architecture of the resulting system, by not allowing for authority or disagreement to be represented as in the original concept.

We do not interpret this pattern as evidence that existing alignment methods are incorrect translations of the traditions they draw on. Computational systems face different constraints and objectives from legal or political institutions, and simplification is often necessary. However, these are not neutral technical simplifications, but are design choices that result in consequences. If such systems make them explicit, and also illustrate how the resulting concept should not be understood as the original, this allows for greater clarity on exactly what normative concept is being upheld. It also makes clear what regions of the design space are still left uncovered, that future research should work towards. We turn next to these underexplored architectures and the design opportunities they suggest

5. Expanding the Normative Design Space

The case studies in Section 4 show how **translation gaps** result when alignment systems only partially implement normative concepts from a tradition, thus making the resulting system represent a different concept in of itself. Here, we discuss a complementary problem of **coverage gaps** i.e., that some sub-spaces of § 3 are underexplored compared to others. We see that there exist regimes in 2 for which only a few technical methods exist, even though traditions in philosophy, political science or legal theory contain relevant that could benefit alignment systems if implemented correctly.

We characterize the gaps as falling into two categories. *Technical gaps* arise for engineering or architectural reasons: when parts of the design space are underexplored because machine learning systems cannot feasibly implement such procedures of conflict management and normative authority. *Mechanism gaps* arise due to infrastructural constraints: when alignment systems omit larger institutional functions such as contesting, appealing, precedent or jurisdiction, by virtue of these normative functions not being implemented by authoritative third-party governance structures. These under-explored concepts serve as structured directions for improving existing methods and building new alternative methods for alignment. Here, we outline some of the largest gaps in the design space (5.1, 5.2), discuss some immediately actionable steps within each dimension, and illustrate research questions that have yet to be explored (**??**).

5.1. Contest

In Figure 2 we see that mechanisms that allow for contesting i.e., preserving disagreement through institutional challenge mechanisms are underexplored. Current alignment methods do incorporate disagreement in terms of real human preferences or opinions during training and evaluation; however most systems resolve and combine the differing judgments into one unified policy. Moreover, the ways in which this resolution occurs often washes away individual or minority view-

Authority × Conflict	Illustrative Architecture	Evaluation Question
Codified × Partition	Multiple constitutions with jurisdictional routing	Can different governance structures provide specifications for contexts?
Case-accumulated × Contest	Precedent-aware alignment with appeals	Can we allow challenges to revise decisions while preserving consistency?
Customary × Partition	Community-specific normative regimes	Can distinct norms coexist without conflicting cross-context application?
Dispositional × Partition	Role-specific agents with learned normative dispositions	Can role-based commitments be learned, maintained and activated across contexts?
Procedural × Contest	Deliberation with persistent dissent	Does dissent meaningfully influence subsequent decisions at individual and community levels?

Table 1 | We illustrate a minimal architecture and evaluation criterion for systems that fall into the axes in our 2×2 taxonomy. Examples are meant to be hypotheses for future systems, rather than claims about which systems might be normatively preferred. Figure 3 highlights the axes that are currently lacking.

points, thus doing away with any artificats of real disagreement in the resulting system. Moreover, institutional mechanisms for this preservation of conflict do not exist. One partial analogue is the process of red-teaming systems, where opposing parties (e.g., human evaluators) discover failures that might exist in a trained model. However, the findings of these are generally incorporated into subsequent updates to model training rather than maintained as an evolving mechanism of ongoing contestation.

Methods that preserve disagreement and the ability to contest decisions should allow for governing principles or model outputs to be challenged, and for these challenge mechanisms to influence future behavior of models through learned mechanisms. This raises the need for technical innovations, where current LLM training should incorporate information from contested decisions in a separate pathway to inform its decision making. This also raises insitutional mechanisms needed to answer questions about what parties should be allowed to raise challenges, at what point in the alignment pipeline these should exist and how competing claims should be evaluated.

5.2. Partition

We see that sub-spaces under the partition dimension are also underexplored—normative orders to accommodate disagreement and keep them as separate, self-contained sub-systems do not exist. In political institutions, such concepts of partition are implemented in mechanisms such as jurisdiction and choice-of-law rules that serve as deciding factors of what normative regime should govern a particular case. Current alignment systems sometimes maintain different normative regimes at different parts of the pipeline: for example, at pretraining stages, constitution-like rules can be incorporated into model training to enforce certain norms, whereas a different set of rules that are more important to safety, or behavior, are incorporated at later stages of training. However, this partition is not systematic, nor is it comprehensive over the types of norms we might requires, and it does not contain jurisdiction rules to combine outputs of norms. Future alignment systems

could maintain multiple constitutions, cultural or community-specific norms, contextual or domain-specific policies, together with explicitly defined mechanisms that select from or govern outputs of each set of norms. This sort of partition makes the boundaries between the sets of norms, as well as authorative power over these norms clear. Technical implementations of this will require clear interaction of model components across stages of the pipeline; while institutional structures will need to answer questions about overlapping authority, assigning jurisdiction and dealing with conflicts between normative orders.

5.3. Addressing Coverage Gaps

Our observation on coverage gaps also connects to translation gaps. Concepts omitted during translation frequently resemble the concepts that distinguish contest and partition regimes: appeal, amendment, persistent dissent, jurisdiction, competing sources of authority, and procedures for revisiting prior decisions. The sparsity of these regions may therefore reflect not an absence of normative resources for designing such systems, but the selective translation of those resources into alignment to date. However, we believe that the lack of existing work here, does not mean that these directions are fruitless. Table **??** illustrates how under-explored sub-spaces have important evaluation question and potential technical architectures. The examples we provide are not meant to be exhaustive proposals, or even claims that these systems are absent from existing pipelines, rather, they show how focusing on a certain sub-space can allow us to generate hypotheses of future architectures that recombine normative mechanisms that are otherwise studied separately. These proposals illustrate the value of this normative design space: that examining the gaps in alignment could allow for the designing of systems that solve problems existing normative structures could not solve.

6. Conclusions

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AI alignment encompasses not only mathematically formulating training objectives to steer systems to human values; but also deciding how normative authority and rules should be structured in cases when those values conflict. In this work, we highlight the choices that are made in AI alignment systems that draw from normative philosophy by introducing a two-dimensional taxonomy to organize existing approaches along two axes. These axes represent how that system incorporates normative authority into its process, as well as how normative conflicts are dealt with (or ignored). Our mapping highlights two clear insights: first, that machine learning implementations of normative concepts can suffer from *normative translation*: when only some concepts of the philosophical, political, and legal traditions they draw on are preserved but others are omitted, this leads to consequences of the resulting system representing a different normative concept. Second, some parts of our two-dimensional property space are far more well-populated than others—and methods that focus on the underexplored areas could lead to new breakthroughs in alignment. Our taxonomy is not exhaustive, especially considering the pace at which AI alignment research leads to new breakthroughs. However, in it's current state, it provides a framework for analyzing existing approaches that we do document and making their normative and institutional assumptions explicit. We believe that this view of alignment highlights a broader, and necessary, research agenda—to develop systems that represent, and deal with diverse viewpoints, have norms as well as governance frameworks in place to allow them to persist over time.

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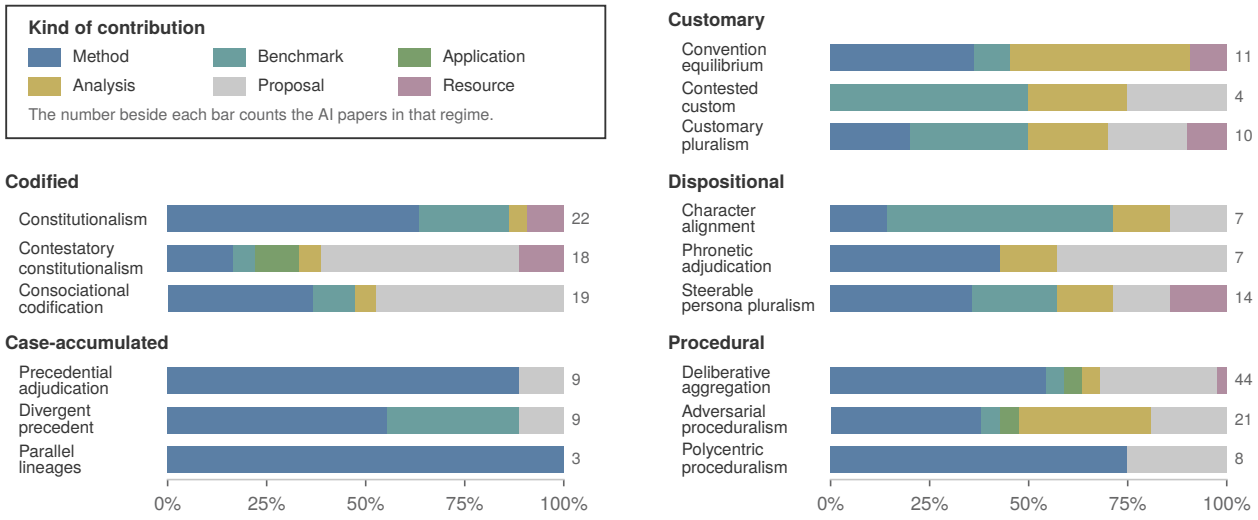


Figure 3 | **Literature Composition.** Breakdown of paper types within each regime. *Methods* specify and demonstrate a mechanism; *Benchmarks* provide evaluation protocols and other resources; *Applications* take an existing method and apply it to a specific subdomain or problem; *Analyses* report findings about existing methods and systems; *Proposals* argue for a design or community action without materializing it; *Resources* are additional artifacts, such as model specs or datasets. [**roma:** this would be great accompanied with the extended related work in the appendix?]

A. Extended Related Work

B. Property Space Construction Details

C. Terminology

Term	Definition	Examples
Normative theory		
Tradition	A body of theory within a discipline.	Constitutionalism; deliberative democracy; legal pluralism
Concept	One constituent of a tradition.	Contestability; rule of recognition; amendment
Institution	A real-world arrangement that realizes a tradition's concepts.	Courts; amendment procedures; elections
Function	The role a concept plays within its institution.	Appeal keeps decisions answerable to those they affect
AI systems		
System	The AI system whose behavior is governed.	A deployed language model assistant
Case	A particular situation in which a system must act.	A request that pits helpfulness against harm avoidance
Mechanism	A technical component.	Reward model; critique-and-revision loop
Method	A concrete technical proposal specifying one or more mechanisms.	Constitutional AI (Bai et al., 2022)
Approach	A family of methods sharing a design.	Debate before an under-informed judge
Regime	A cell of the property space: one locus of normative authority and one disposition of normative conflict.	Contestatory constitutionalism; adversarial proceduralism
Norms		
Norm	Any standard that guides a system's behavior, whatever its form.	Rules; principles; precedents; character traits
Rule	A norm stated explicitly in advance.	"Do not give instructions for building weapons."
Principle	A general norm in a written specification.	"Choose the most helpful and honest response."
Normative translation		
Normative translation	How a method carries over the concepts of a tradition it draws on.	Constitutional AI translating constitutionalism
Preserved / partially preserved / omitted	Whether a method carries over a concept fully, in part, or not at all.	Constitutional AI preserves codification and omits amendment
Addition	Something a method contributes beyond its tradition.	Self-critique by the model
Shortfall	An omission of a concept the method claims to draw on.	A method claiming contestability without any route to appeal
Translation gap	A concept of a tradition that methods drawing on it do not carry over.	Appeal in Constitutional AI
Coverage gap	A regime with few or no technical methods.	Contested custom